Pip saw it. Why did Pip not stop?

Detection is not the same as action.



By the end, diagnose the whole system

- A · Bits to programs to actions
- B · Hardware to packets to trust
- C · Control to constraints to decisions

Same bits. Different meanings.

01000001

65 as an unsigned integer

A in ASCII

Predict the value before decoding

8 bits: 11111101

Unsigned: ?

Two's complement: ?

YOUR TURN · 45 SECONDS

The width and encoding decide the value

Unsigned: 128 + 64 + 32 + 16 + 8 + 4 + 1 = 253

Signed: −128 + 64 + 32 + 16 + 8 + 4 + 1 = −3

An image is a grid of encoded colours

100 × 100 pixels

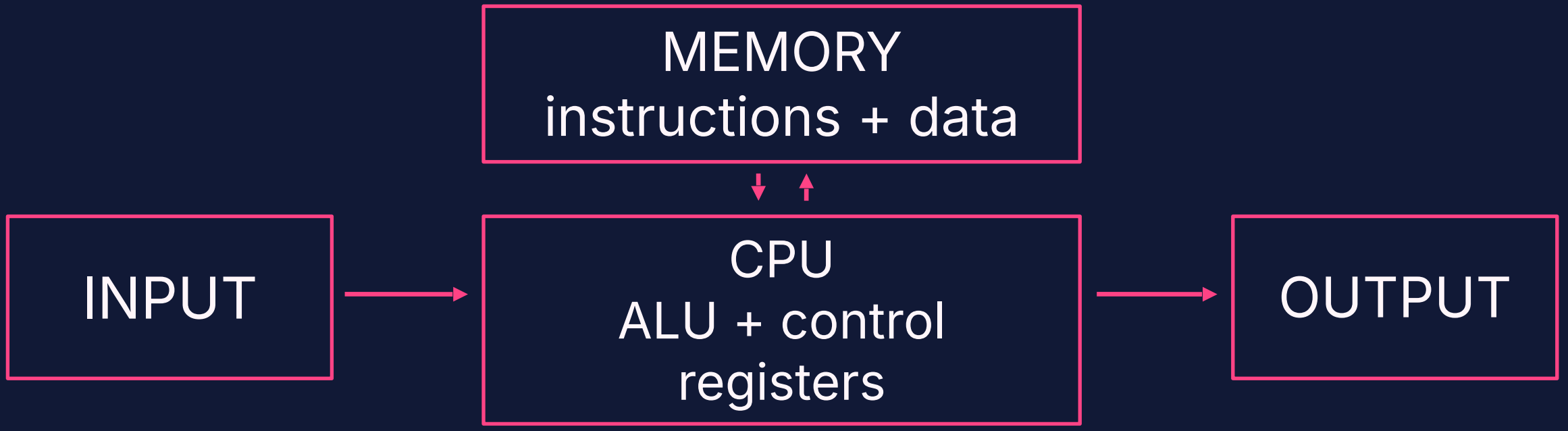
24 bits per pixel

100 × 100 × 24 ÷ 8 = 30,000 bytes

Sampling turns a signal into numbers



Instructions and data share memory



One instruction, three useful questions



Can you trace this tiny program?

Address	Instruction
100	LOAD #3
101	ADD #4
102	STORE 200
103	HALT

YOUR TURN · 60 SECONDS

The result is a state change

Instruction	ACC	memory[200]
LOAD #3	0 to 3	—
ADD #4	3 to 7	—
STORE 200	7 to 7	7
HALT	7 to 7	7

Source code is not a CPU instruction



The operating system shares the machine

- CPU time + memory
- Files + devices
- Protection + coordination

An interrupt is a request for attention



A response can be designed or learned

Reactive rule	If obstacle is near, stop.
Learned policy	Choose actions using experience.

Urgent behaviour can take priority

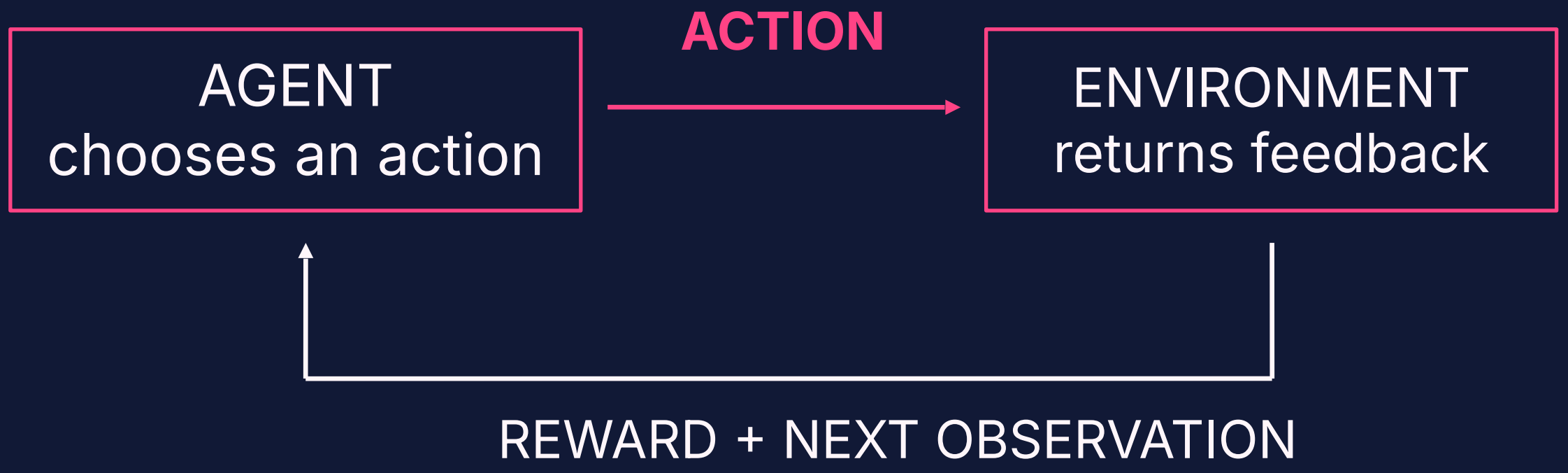
AVOID COLLISION

SEEK A GOAL

EXPLORE

YOUR TURN · 20 SECONDS

Learning closes a feedback loop



Policy chooses. Value estimates.

POLICY	Which action should I take?
VALUE	How much return do I expect?

Try something new—or use what works?

Route A: known reward estimate 3

Route B: barely explored

Which should the agent try?

YOUR TURN · 45 SECONDS

A Q update moves partway to a target

Old Q = 2 · learning rate = 0.5

Reward = 1 · discount = 0.9 · next best Q = 4

Target = 1 + 0.9 × 4 = 4.6

Where does the estimate move?

Old estimate: 2

Target: 4.6

Move halfway: new Q = ?

YOUR TURN · 30 SECONDS

The new estimate is 3.3

Error = 4.6 − 2 = 2.6

Change = 0.5 × 2.6 = 1.3

New Q = 2 + 1.3 = 3.3

An MDP makes assumptions explicit

- STATES + ACTIONS
- TRANSITIONS + REWARDS
- DISCOUNT + MARKOV ASSUMPTION

Training and deployment have different jobs

TRAIN	Collect experience; update the policy.
DEPLOY	Use a policy; monitor its behavior.

Session A checkpoint: explain an action

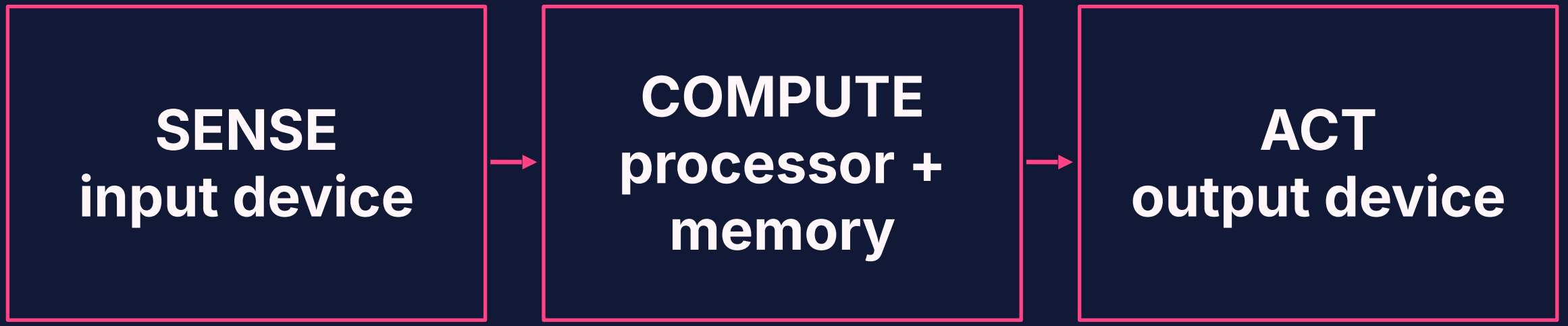
Bits need an interpretation.

Programs change machine state.

A policy chooses; feedback can update it.

YOUR TURN · 60 SECONDS

A good decision still needs a body



Memory trades capacity against access time



Closer: lower latency · farther: more capacity

Keep data after the power disappears

RAM	Working state; generally volatile.
FLASH / SSD	Non-volatile electronic storage.
HDD / OPTICAL	Magnetic / laser-readable storage.

Pipelining overlaps different instructions

Cycle	1	2	3	4	5	6
I1	F	D	E	W	—	—
I2	—	F	D	E	W	—
I3	—	—	F	D	E	W

F Fetch · D Decode · E Execute · W Write

YOUR TURN · 30 SECONDS

Parallel hardware is useful only when work fits

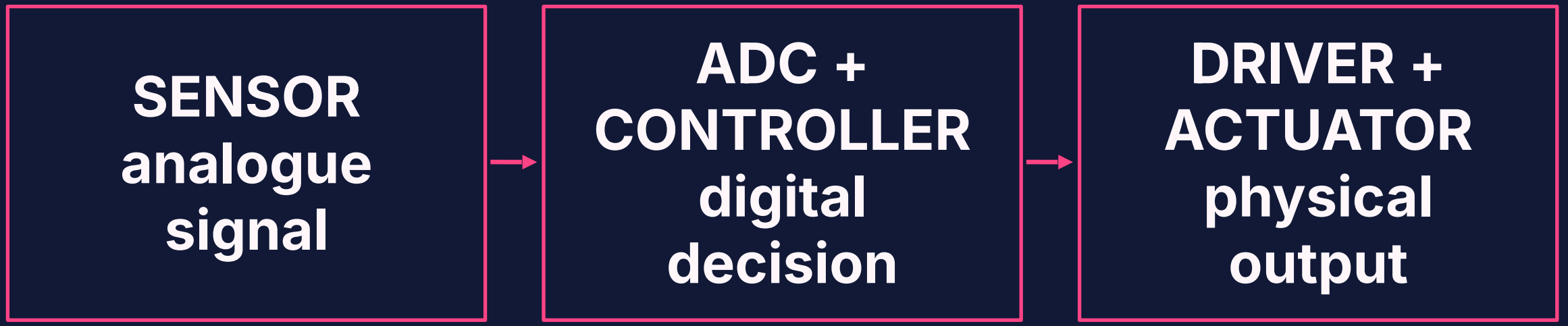
CPU	Flexible control and varied work.
GPU / NPU	Many suitable operations in parallel.

A logic gate makes a precise choice

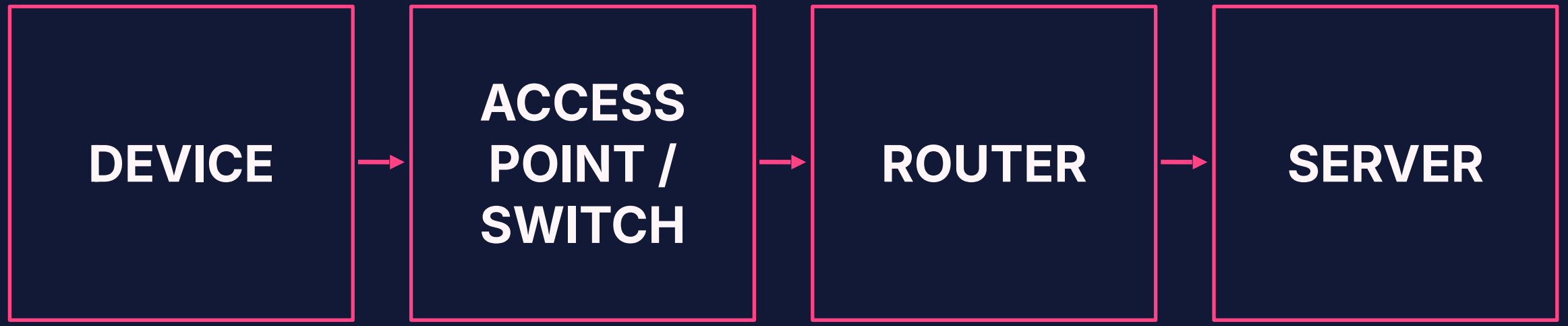
Enabled	Safe	AND output
0	0	0
0	1	0
1	0	0
1	1	1

YOUR TURN · 20 SECONDS

The physical world needs an interface



A message crosses several different jobs



Local hop to network boundary to remote service

Who requests—and who provides?

CLIENT / SERVER	Requester and service provider.
PEER TO PEER	Participants can do both jobs.

Name resolution is not page delivery



Layering separates responsibilities

APPLICATION · message meaning

TRANSPORT · process to process

INTERNET · packet routing

LINK · the local hop

Reliable delivery is not a deadline guarantee

TCP	Ordered reliable byte stream.
UDP	Datagrams without that guarantee.

Shared capacity creates waiting



Capacity is not responsiveness

BANDWIDTH · available capacity

THROUGHPUT · achieved transfer rate

LATENCY / JITTER · delay / variation

How far does the robot move while waiting?

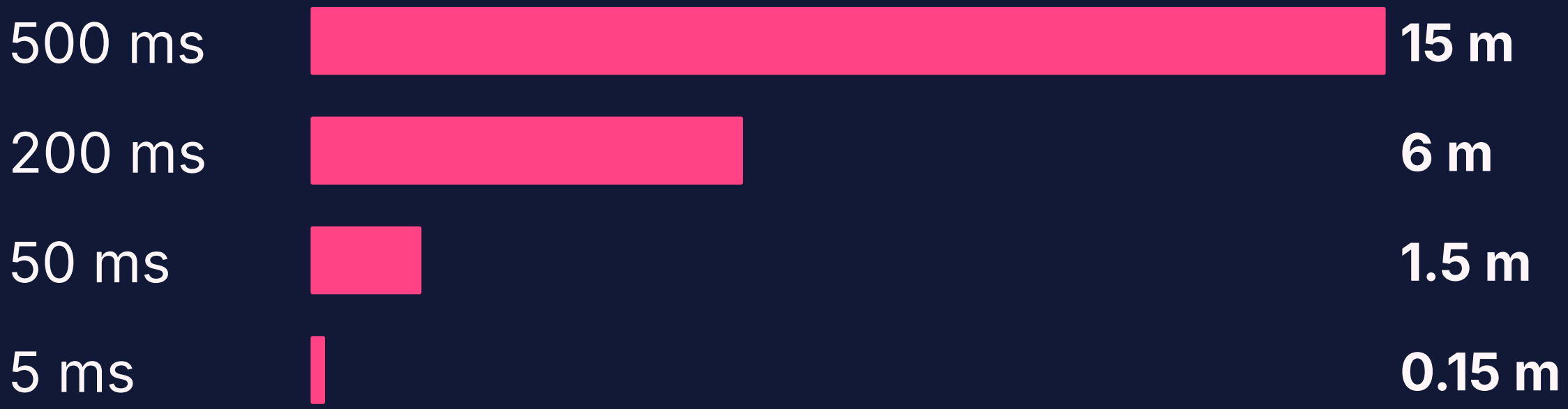
Speed = 30 metres / second

Delay = 200 milliseconds

Distance during delay = ?

YOUR TURN · 45 SECONDS

A short delay can have a physical consequence



At 30 m/s · distance during delay only

Correct delivery is not correct meaning

VALIDATION	Does this value obey the rules?
VERIFICATION	Does it match a source or expected copy?

Protect secrecy, identity and integrity

- ENCRYPTION · keep data confidential
- SIGNATURE / MAC · detect changes, authenticate
- CERTIFICATE · bind an identity to a key

Secure transport still needs safe application logic



Session B checkpoint: useful, not merely delivered

Hardware limits execution.

Networks add delay and uncertainty.

A secure packet can still be stale.

YOUR TURN · 60 SECONDS

A smart system is a chain of responsibilities



Irrigation: what should happen when data is missing?

Moisture report is old.

The valve can flood the bed.

Keep watering—or pause and alert?

YOUR TURN · 45 SECONDS

A deadline is part of the result

CORRECT VALUE, TOO LATE	May be unusable for control.
CORRECT VALUE, ON TIME	Can inform the intended decision.

A control message needs context

Device	robot-7
Units / policy	metres / version 3
Sequence / age	42 / 50 ms

Which messages should the robot accept?

Message	Sequence	Age
A	42	50 ms
B	43	600 ms
C	42	60 ms

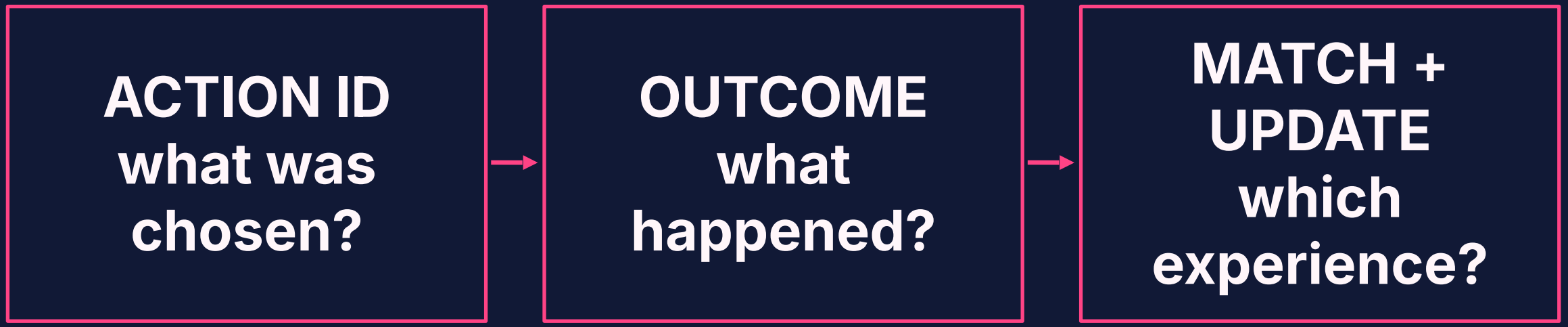
Start at sequence 41 · age limit 200 ms

YOUR TURN · 60 SECONDS

Accept once. Reject stale and duplicate commands.

Message	Decision	Last sequence
A · 42 · 50 ms	ACCEPT	41 to 42
B · 43 · 600 ms	STALE	42 to 42
C · 42 · 60 ms	DUPLICATE	42 to 42

Delayed feedback must belong to the right action



People need an understandable system state

- SHOW · what the system knows
- EXPLAIN · uncertainty and limits
- ALLOW · oversight and recovery

A faster system can still be a worse system

MEASURE PERFORMANCE	Timing, accuracy, energy, reliability.
MEASURE IMPACT	Privacy, fairness, access, accountability.

Four physical budgets constrain every design

- TIME
- ENERGY + HEAT
- MEMORY + COMPUTE

Bad inputs can look perfectly plausible

SENSOR DRIFT	Measurements gradually change.
DISTRIBUTION SHIFT	Deployment differs from training.

More devices create coordination work

- SCALABILITY · handle growth
- INTEROPERABILITY · share meaning
- RESILIENCE · handle disruption

Predicting an outcome is not explaining behavior

PREDICTIVE ANALYTICS	Forecast a future outcome.
BEHAVIOURAL MODELLING	Represent actions and interactions.

Final design challenge: rescue Pip

Local control: 5 ms loop

Remote service: variable 500 ms delay

Place stop, training and fleet planning.

Keep the urgent loop local; coordinate when useful



The opening mystery has five possible failure points

Representation + execution

Policy + hardware

Network timing + application checks

YOUR TURN · 30 SECONDS

Continue with the matching learning resources

- CH 1–2 · trace a program; update a Q value
- CH 3–4 · trace a pipeline; explain delay
- CH 5–6 · validate a packet; justify an architecture

Sources, scope and production credit

- BLUE IMPULSE · Stage 2 · PPT V1.0
- Original editable diagrams + checked teaching models
- Full references and AI evidence included in this package